

Resilience Analytics for Interdependent Power and Water Distribution Systems

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Abstract—The occurrence of large-scale outages in power distribution systems (PDSs) caused, for instance, by extreme weather events has raised concerns on the cascading impact of the power outages on the resilience of other critical infrastructures, including water distribution systems (WDSs), whose healthy operation highly relies on the availability of power from PDSs. This paper proposes an analytics model for quantifying the interdependence between the resilience of power and water distribution systems. The proposed model first performs spatio-temporal outage analysis on the PDS and WDS to determine the PDS power serving capability and the resulting cascading impact on the WDS water serving capability during the occurrence of disruptions. The resilience interdependence is then determined by a proposed set of six metrics calculated using the power and water serving capability profiles, which trace and quantify the cascading impacts of power outages in the operation of WDSs during the degradation and recovery stages. The proposed analytics model is implemented on the IEEE 33-bus power distribution system supplying power for pumping stations of a 15-node test WDS, considering multiple cases of water storage capacity, single and multiple outage scenarios, and availability of distributed generation. The numerical results demonstrate that the proposed metrics track the temporal and spatial interdependence between the resilience of power and water distribution systems, providing a mechanism to identify weak spots, and assess options to enhance the resilience of the critical water infrastructure.

Index Terms—Water-power nexus, water distribution system, power distribution system, resilience metric, outage analysis.

NOMENCLATURE

A. Sets and Indices

s	Index of outage scenarios.
i, j	Indices of power buses.
ij	Index of the line connecting buses i and j .
t	Index of time steps.
u	Index of distributed generation units.

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e	Index of ES systems.
n	Index of water nodes.
B	Number of power buses.
N	Number of water nodes.
P	Number of water pipelines.
M	Number of water pumps.
T	Number of time steps in operation horizon.
S	Set of outage scenarios.
$\mathcal{T}, \mathcal{T}'$	Set of time steps in operation horizon.
$\mathcal{B}$	Set of power buses.
$\mathcal{L}$	Set of power lines.
$\mathcal{L}_s$	Subset of outage lines in scenario s .
$\mathcal{U}$	Set of distributed generation units.
$\mathcal{U}_i$	Subset of distributed generation units connected to bus i .
$\mathcal{E}$	Set of ES systems.
$\mathcal{E}_i$	Subset of ES systems connected to bus i .
$\mathcal{N}$	Set of water nodes.

B. Parameters

Δt	Time step.
$EC_{i,t}$	Value of lost load at bus i at time t .
$p_{i,t,s}^D, q_{i,t,s}^D$	Active/reactive power demand at bus i in scenario s during time t .
$g^{ij}, b^{ij}, b_{sh}^{ij}$	Conductance, susceptance, and shunt admittance of line section ij .
$\alpha_{t,s}^{ij}$	Status of line ij at time t during scenario s .
$\overline{v_i}, \underline{v_i}$	Min/max voltage magnitude at bus i .
$\overline{S^{ij}}$	Maximum capacity of power line ij .
$\overline{p_u^{DG}}$	Maximum active power output of distributed generation unit u .
$\overline{q_u^{DG}}, \underline{q_u^{DG}}$	Min/max reactive power output of distributed generation unit u .
$\overline{F_u}$	Available fuel of distributed generation unit u .
η_e^c, η_e^d	Charging/discharging efficiency rates of ES system e .
$\overline{E_e}, \underline{E_e}$	Min/max ES capacity of system e .
$\overline{B_e}$	Maximum power input/output of ES system e .
$\overline{q_e^{ES}}, \underline{q_e^{ES}}$	Min/max reactive power output of ES system e .
$\overline{WC_n}$	Unit cost of water not supplied at node n .
$W_{n,t,s}^D$	Water demand at node n , time t during scenario s .
$\mathbf{A}$	$P \times N$ pipeline-to-node incidence matrix.
$\mathbf{\Lambda}$	$U \times N$ matrix mapping water tanks to water nodes.

$\mathbf{1}_N, \mathbf{1}_M$	N/M -dimensional vector of ones.
Δ	$U \times U$ diagonal matrix of water tank area.
$\bar{\mathbf{V}}, \mathbf{V}$	$U \times 1$ vector of max/min tank capacity.
$\mathbf{V}^{init}$	$U \times 1$ vector of initial water volume.
$\bar{\mathbf{C}}, \bar{\mathbf{D}}$	$U \times 1$ vector of maximum tank input/output capacity.
g	Gravity.
$\mathbf{G}$	$P \times P$ diagonal matrix of pipe friction coefficients.
$\bar{\mathbf{Q}}$	$P \times 1$ vector of maximum water flow.
$\underline{\mathbf{H}}, \bar{\mathbf{H}}$	$N \times 1$ vector of min/max pressure head.
a, b, c	Characteristic parameters of pumps.
$\mathbf{B}$	$M \times P$ matrix mapping pumps to arcs.
ρ	Water density.
η_{pump}	Pump efficiency.
$\mathbf{P}^W$	$M \times 1$ vector of maximum pump power consumption.
$\mathbf{I}_{t,s}$	$M \times 1$ binary vector of power outage status of pumps.
$\bar{\mathbf{P}}^B$	$M \times 1$ vector of maximum backup generator power output.
$\bar{\mathbf{F}}^B$	$M \times 1$ vector of backup generator available fuel.

C. Variables

$\gamma_{i,t,s}^P$	Ratio of power served at bus i in scenario s during time t .
$p_{t,s}^{grid}, q_{t,s}^{grid}$	Active/reactive power supplied by the bulk power grid at time t of scenario s .
$p_{t,s}^{ij}, q_{t,s}^{ij}$	Active/reactive power flow from bus i to bus j at time t of scenario s .
$p_{u,t,s}^{DG}, q_{u,t,s}^{DG}$	Active/reactive power supplied by the distributed generation unit u at time t .
$D_{e,t,s}^{ES}$	Discharging rates of the ES system e connected to bus i of scenario s .
$C_{e,t,s}^{ES}$	Charging rates of the ES system e connected to bus i of scenario s .
$q_{e,t,s}^{ES}$	Reactive power supplied by the ES system e at time t .
$v_{i,t,s}$	Voltage magnitude at bus i at time t of scenario s .
$\theta_{i,t,s}$	Voltage phase angle at bus i at time t of scenario s .
$E_{e,t,s}$	State-of-charge of ES system e at time t of scenario s .
$\gamma_{n,t,s}^W$	Ratio of water served at water node n in scenario s at time t .
$\mathbf{W}_{t,s}^R$	$N \times 1$ vector of water injections from reservoirs to nodes.
$\mathbf{Q}_{t,s}$	$P \times 1$ vector of pipeline water flow.
$\mathbf{C}_{t,s}, \mathbf{D}_{t,s}$	$U \times 1$ vector of water tank charging/discharging flow.
$\mathbf{\Gamma}_{t,s}^W$	$N \times 1$ vector of ratio of water served.
$\mathbf{V}_{t,s}$	$U \times 1$ vector of tank volume.
$\mathbf{H}_{t,s}$	$N \times 1$ vector of pressure head.
Θ	$R \times N$ matrix mapping water reservoirs to water nodes.
$\mathbf{Z}$	$R \times 1$ vector of reservoirs' elevations.

$\Delta \mathbf{H}_{t,s}^P$	$M \times 1$ vector of head rise created by pumps.
$\mathbf{\Omega}_{t,s}$	$M \times 1$ vector of pump speeds.
$\mathbf{P}_{t,s}^W$	$M \times 1$ vector of power injections to the WDS.
$\mathbf{P}_{t,s}^B$	$M \times 1$ vector of power injections from backup generators.
$\mathbf{Q}_{t,s}^P$	$M \times 1$ vector of water flows in pumps.

I. INTRODUCTION

A. Background and Literature Review

The interdependence between power distribution systems (PDSs) and water distribution systems (WDSs) has received growing attention in recent years, focusing on opportunities for coordinated operation to improve the performance of both systems [1], [2]. The other side of this interdependence, however, is related to the vulnerability to outages that cascade from power grid into the water infrastructure. The PDSs are exposed to a wide variety of faults caused by equipment failures or external events [3]–[6], which can cause subsequent cascading outages in WDSs that are powered by the PDS. The vulnerability of PDSs to such events has been experienced during multiple, recent large-scale catastrophic events, Superstorm Sandy in 2012 and Hurricane Maria in 2017, which have disrupted the operation of other infrastructures dependent on power supply, including water and transportation systems [7]–[10]. In particular, the interaction between power and water distribution systems is critical in case of disasters, since their extended interruption can represent a life-threatening risk for vulnerable communities, as occurred in Texas during the 2021 winter storm [11].

Despite the vulnerabilities of power and water infrastructures being well known, actions aimed at enhancing the resilience of PDSs and WDSs are based on metrics that treat these systems as independent, neglecting the effect of their interdependent operation. Strategies for enhancing resilience of PDSs include splitting the grid into microgrids for self-healing operation [5], utilizing distributed resources for local load restoration [12], [13], distribution line hardening [14], and grid reconfiguration to maximize the power supply to the affected demand [15], where resilience is associated with the amount of load shed during recovery. Alternatively, the resilience assessment is achieved based on a performance curve that assesses the evolution of a resilience indicator in time, the triangle curve or the trapezoid curve [16], [17]. The trapezoid curve is utilized in [18] to formulate five metrics which describe the phases experienced by a critical infrastructure during the disaster and its restoration. In addition, six probabilistic metrics are introduced in [19] to capture the response of automated devices in the grid during the fault isolation and restoration. These metrics are then used in [20] to assess the resilience of the grid against a specific type of natural disaster. Furthermore, a metric to quantify the system resilience to a time-evolving event is presented in [21] to consider the risk to extreme temperatures and water scarcity. Besides, the quantification of resilience based on load shedding is extended to involve a component criticality ranking in [22] to

support operators during cost-benefit analyses. Moreover, several resilience frameworks for microgrids based on performance curves are found in literature, including metrics to quantify vulnerability, degradation, and restoration efficiency [23], resilience quantified as the reciprocal of the performance loss [24], a unified resilience metric to integrate the network constraints [25], and metrics based on the energy supplied compared to the total demand [26], [27]. The benefits of these solutions, however, are solely analyzed for load restoration in the PDS and not for the interdependent systems. It is discussed in [28] that PDS restoration actions should involve ensuring critical load supply to other infrastructures that are dependent on power supply for operation.

Similar to PDSs, the hardening actions for WDSs, which include upgrading pump stations, increasing tank capacity, pipe retrofitting, network expansion, and adaptive operation for pressure management, are defined according to metrics that only consider the inherent characteristics of the pipelines and the WDS topology, neglecting the interactions with the power infrastructure that supplies power for operation of the WDS [29]–[33]. The resilience quantification in WDSs has followed two major methods. On one side, the metrics use the definition introduced in [34], where resilience is quantified as the ability to return to a non-failure state after a failure has occurred. On the other hand, resilience is quantified by the ability of WDS to deliver water based on the available pressure head surplus at each node, the amount of pressure head above the minimum required for water distribution [35]. Additional metrics have been proposed in the literature based on these methods [29], [30], which are solely developed for a stand-alone WDS.

In general, the resilience quantification in critical infrastructures is associated with the amount of demand supplied during and after the event. A number of resilience metrics is developed by modeling the infrastructures as graphs, where resilience is defined using the ratio between demand satisfaction and total demand [36] or the weighted sum of the individual performance of each infrastructure [37]. While this approach offers a modeling alternative for any interdependent infrastructure, the resilience assessment requires developing methods that consider specific constraints of each system, which cause inaccuracies when neglected [38]. In particular, metrics to quantify resilience of interdependent power-gas networks are formulated in [39], [40], which are respectively obtained based on the cost of the curtailed energy demand and the convex linear combination of each infrastructure performance based on budget. In power and transportation infrastructures, where power outages cause traffic light de-energization, resilience is quantified as a function of the cost of power curtailed and the cost of travel times after power interruptions for traffic lights [10] or as the ratio between the travel time before and after the event [41].

The interdependent nature of the PDS and WDS operation has been scarcely considered in metrics for resilience assessment. A study based on data retrieved from the 1994 Northridge earthquake was conducted in [42], where pumps stations were identified as vulnerable elements during the power supply interruption. A framework for enhancing resilience in interdependent power-water infrastructures is developed in [43], where the

resilience is quantified based on the number of users with access to electricity and water supply after a disaster. On the other hand, a metric is proposed in [44] to aggregate several independent and interdependent performance functions of a power transmission grid and a WDS, weighted in a single indicator based on a WDS sensitivity analysis. Despite the existing methodologies, there is a lack of mechanisms to model and quantify the interdependence between the resilience of power and water distribution systems during the response and recovery after disasters, which are key tools to identify vulnerabilities affecting the infrastructures during the high-impact events.

B. Contribution and Paper Structure

This paper proposes an analytics model for quantifying the interdependence between the resilience of power and water distribution systems. The proposed model quantifies the cascading impact of power outage in PDS that is propagated into the WDS by disrupting the operation of pumping stations, which would subsequently cause a water shortage in the WDS. Different to methodologies based on using a generic metric to assess the resilience of power systems with water and other infrastructures, the proposed approach quantifies the interdependence between the resilience of power and water infrastructures by tracking how the PDS response during a disruption is cascaded into the WDS operation, considering different stages of the operation and performance of the systems after outages. The proposed approach first determines the average power and water serving capabilities resulting from the spatio-temporal outage scenarios emulating the impacts of an external event. The importance of generating spatio-temporal outage scenarios is to include the impact of the outage location on the WDS resilience. Also, the assessment is not limited to a single scenario and instead aims at considering multiple scenarios representing different outcomes from the external event.

Furthermore, instead of defining a single unified resilience indicator, a set of six indicators are proposed, which are carefully formulated to quantify effects on the WDS serving capability caused by interruptions in the power supply during the outage affecting the PDS serving capability. More specifically, the six indicators aim at quantifying relative effects of the PDS outages on the WDS operation and resilience, evidenced in different stages of the serving capability curves associated with the effect of the PDS degradation on the WDS degradation, the effect of the PDS restoration on the WDS recovery, the ratio of supplied water with respect to amount of power in the PDS, the similarities between the power and water serving capabilities, and the time differences between the power and water disruption and restoration. Hence, the metrics aim at providing sensitivity to WDS vulnerabilities intensified by its interdependence with PDS and expect to motivate utilities from different infrastructures to switch paradigms from an isolated and independent planning and operation to a comprehensive framework where efforts are coordinated to make their systems more resilient. The contributions of the paper are summarized as follows:

- An analytics framework is developed and evaluated, which performs the generation of spatio-temporal outage scenarios, the evaluation of these scenarios in the PDS using a second-order cone programming (SOCP) model, and the assessment of the cascading effect of power outages into the WDS operation during the disaster.
- A cascading water shortage analysis model is proposed to optimize the operation of healthy pumping stations and water storage tanks in WDS and minimize the volume and duration of water shortage caused by a power outage in pumping stations in a given PDS outage scenario.
- A set of six metrics is proposed to quantify the interdependence between the resilience of power and water distribution systems. The proposed metrics quantify the dependency of WDS resilience to PDS in terms of degradation, recovery, and total loss of service and measure the temporal delay between power outage in PDS and water shortage in WDS. The proposed metrics enable studying and quantifying the impacts of PDS vulnerabilities, as well as the benefits from measures taken to improve the resilience of PDS on the resilience of WDS.

The rest of the paper is organized as follows: Section II presents an overview of the proposed model and describes the cascading effect of PDS disruptions to the WDS operation. The formulation of the mathematical models for power outage analysis in PDS and cascading water shortage analysis in WDS are presented in Section III. The proposed metrics to quantify the interdependent resilience of PDS and WDS are defined in Section IV. The numerical results of implementing the proposed model on the IEEE 33-bus test PDS and a 15-node test WDS are presented in Section V. Conclusions are presented in Section VI.

II. OVERVIEW OF THE PROPOSED MODEL

A reliable power supply is critical for the operation of energy-intensive processes in WDSs, including water withdrawal, treatment, and pumping, which are supplied by PDSs [2], [45]. According to [46], almost all of the water treatment facilities in the United States are dependent on an external power supply to maintain their core operation. The occurrence of faults in the PDS may result in a power outage in the water facility and disrupt the operation of WDS. The mitigation of power outage risk on water treatment and water distribution facilities relies mostly on the use of fuel-based backup power generators, including diesel, natural gas, propane, and gasoline generators. Stationary engines and portable generators are used depending on the amount of power to be supplied. In case of large demand, stationary engines provide uninterrupted electricity to supply the demand and power pumps, while less expensive portable generators are preferred to supply facilities with low power consumption. In the United States, these portable generators can be owned by the water facility or borrowed from other utilities before a natural disaster through the Water/Wastewater Agency Response Network (WARN) to ensure the backup power requirements are met during the event [47], [48]. Although 88% of the treatment facilities are equipped with some form of backup generation,

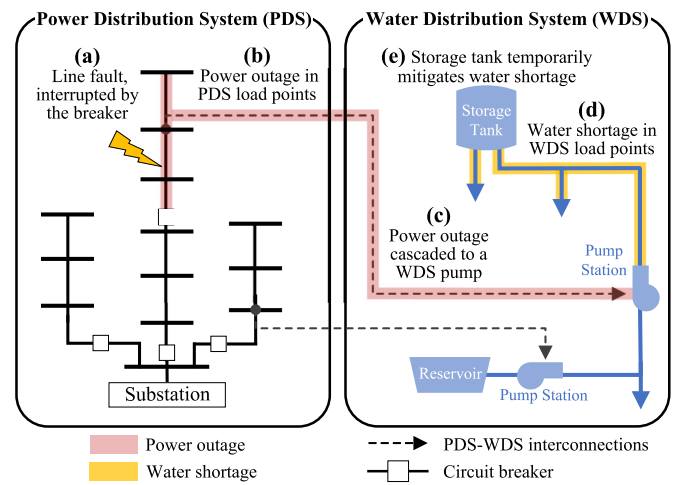


Fig. 1. Propagation of power outage in PDS to water shortage in WDS.

including distributed generators, portable generators, and uninterruptible power supply (UPS), their capacity is typically not big enough to fully supply the power demand of the water facility during external power outages. The use of fuel-based emergency generators is regulated by the U.S. Environmental Protection Agency (EPA) according to the Clean Air Act [49], which limits the available power at water facilities to sustain the operation during the emergency. In developing countries with rural WDSs and irrigation districts, the availability of backup power is even more restricted, due to the budget limitations and the risk of soil contamination from fuels [50].

An illustrative example of fault propagation from PDS to WDS is shown in Fig. 1, which starts by a line fault in PDS and subsequent operation of protection devices and circuit breakers in (a). This results in the isolation of line sections downstream of the breaker and power outage in the PDS load points in (b) in Fig. 1. The power outage in the affected PDS buses interrupts the power supply to the pumping station in (c), which in turn limits the WDS ability to ensure adequate water pressure for distribution, leading to potential water supply shortages downstream of the pump in (d). The water shortage may be compensated temporarily by the water stored in the tank in (e). This propagation of power outage from PDS to disruption and water shortage in WDS is defined in this paper as cascading outage between PDS and WDS, which is dependent on the WDS configuration, the available water storage, and the interconnection between the systems.

In order to assess the resilience of interdependent power and water distribution systems given these cascading outages, this paper proposes a model for resilience analytics of interdependent power and water distribution systems. The proposed model quantifies the resilience of both systems using a set of six resilience metrics, which quantify the impacts of PDS outage scenarios in WDS operation according to the temporal evolution of water shortage during power outages. The interdependence of power systems to water systems at power transmission level is well-known, since water is required for hydropower generation

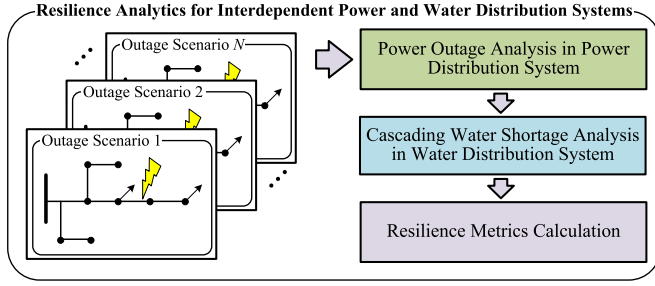


Fig. 2. The proposed resilience analytics model for interdependent power and water distribution systems.

and the cooling process of thermal power plants. At power distribution level, power generation technologies (small thermal units or renewable resources) do not typically rely on water inflow for cooling or power generation. However, emerging technologies, including small-scale pumped-storage hydropower [51], can potentially increase the interdependence of power distribution systems to water distribution systems, which can be used to enhance the resilience of both infrastructures. This dependency, however, is not explicitly considered in this paper for the metrics development. The structure of the proposed resilience analytics model is illustrated in Fig. 2. The proposed model first performs an outage analysis to determine the actions to be taken by the PDS to minimize the demand curtailment of each system, in response to the spatio-temporal outage scenarios that model the impact of random external events on the PDS operation. The actions taken by PDS operator include the power transacted with the main grid and the power exchange with distributed generators and energy storage (ES) systems. The proposed model, then, determines the cascading impacts of power outage on the operation of WDS, in terms of the resulting water shortage. Given the results of power outage and water shortage in different scenarios, the interdependence between the resilience of the PDS and WDS is assessed using the proposed set of six resilience metrics. The metrics identify temporal PDS and WDS dependencies during the system degradation and recovery stages, as well as the spatial features associated with the fault location and water curtailments. The proposed models for outage analysis in power and water distribution systems are presented in Section III, and the proposed resilience metrics are introduced in Section IV.

III. OUTAGE ANALYSIS IN INTERDEPENDENT POWER AND WATER DISTRIBUTION SYSTEMS

This section presents the proposed models for outage analysis in interdependent power and water distribution systems. In the following, we first introduce the proposed spatio-temporal outage scenario generation and evaluation setup, followed by the mathematical formulation of models for power outage analysis in power distribution and the cascading water shortage analysis in water distribution system under the scenarios.

A. Spatio-Temporal Outage Scenario Model

Resilience is assessed by evaluating scenarios that represent the uncertain behavior of external events (such as natural disasters) and the response of the PDS subject to disruptions. Let the outage scenario $s \in \mathcal{S}$ include outage of subset $\mathcal{L}_s = \{1, \dots, L_s\}$ of set of all lines $\mathcal{L}$ in PDS, where each line outage $l \in \mathcal{L}_s$ is described by a triplet $(l, t_{s,l}^d, t_{s,l}^r)$ with $t_{s,l}^d$ and $t_{s,l}^r$ representing the outage time and restoration time of line section l in scenario s . The outage time $t_{s,l}^d$ is determined according to the intensity of the external event, e.g., hurricane, and the fragility of the line. The restoration time $t_{s,l}^r$ is dependent on the time it takes by repair crew to repair and restore the damaged line. Each outage scenario s , therefore, represents up to $2L_s$ number of outage and restoration events associated with L_s line outages (the number of events could be less if the outage/repair of lines coincide). For the scenario evaluation, the operation horizon $\mathcal{T}$ is divided into sequential sub-horizons defined by the number of topology transitions induced by the generated spatio-temporal scenario. A topology transition at time $t \in \mathcal{T} \setminus \{1\}$ refers to a change in the health status of power distribution lines between time $t - 1$ and t . Based on the outage and restoration times of the damaged lines, a set of $R_s + 1$ sub-horizons are generated for each scenario $s \in \mathcal{S}$, where R_s is the number of topology transitions resulting from the outage/restoration events in the scenario. Each sub-horizon, which starts at the time of the topology transition and is initialized with the previous sub-horizon results, may represent a normal condition where all power distribution lines are operating (pre-contingency and restored states) or faulted states with single or multiple line outages. The solution of each sub-horizon corresponds to the reallocation of power and water stored resources, given the new topology status after the outage/restoration actions.

B. Power Distribution Outage Analysis Model

Let each sub-horizon $\mathcal{T}'$ in outage scenario s be divided into time periods of same length Δt . The objective of the proposed PDS outage analysis model is to minimize the cost of energy not supplied in scenario s during sub-horizon $\mathcal{T}'$ as described in Section III-A, which is formulated as follows:

$$\min \sum_{t \in \mathcal{T}'} \sum_{i \in \mathcal{B}} EC_{i,t} \times (1 - \gamma_{i,t,s}^P) p_{i,t,s}^D \Delta t, \quad (1)$$

where $EC_{i,t}$ is the unit cost of energy not supplied at bus i at time t , $p_{i,t,s}^D$ is the active power demand at bus i during time t , and $\gamma_{i,t,s}^P$ is the ratio of power served at bus i in scenario s during time t . The objective function is subject to constraints (2)–(21), which are defined for $t \in \mathcal{T}'$ except where specified.

1) *Power Balance Constraints*: The active and reactive power supplied to PDS by the bulk power grid at the substation bus, $p_{t,s}^{grid}, q_{t,s}^{grid}$, are equal to the summation of the power flows of distribution lines connected to the main substation bus, given by (2), where $p_{t,s}^{1j}, q_{t,s}^{1j}$ are the active and reactive power flow between the substation bus 1 and the set of buses connected to the substation bus 1, respectively. Power balance at distribution buses is given by (3)–(4), which integrates power injections from available set $\mathcal{U}_i$ of distributed generation (DG) and set $\mathcal{E}_i$ of ES

systems at bus i , where $p_{u,t,s}^{DG}, q_{u,t,s}^{DG}$ represent the active and reactive power supplied by DG unit u connected to bus i at time t ; $D_{e,t,s}^{ES}, C_{e,t,s}^{ES}$ are the discharging and charging rates of ES system e connected to bus i at time t ; $q_{e,t,s}^{ES}$ is the reactive power supplied by ES system e connected to bus i at time t ; $p_{i,t,s}^D, q_{i,t,s}^D$ are the active and reactive power demand of bus i at time t ; and $p_{t,s}^{ij}, q_{t,s}^{ij}$ are the active and reactive power flows from bus i to buses j that are connected to bus i shown by $\mathcal{B}_i$.

$$p_{t,s}^{grid} = \sum_{j \in \mathcal{B}_1} p_{t,s}^{1j}, \quad q_{t,s}^{grid} = \sum_{j \in \mathcal{B}_1} q_{t,s}^{1j}, \quad (2)$$

$$\sum_{u \in \mathcal{U}_i} p_{u,t,s}^{DG} + \sum_{e \in \mathcal{E}_i} (D_{e,t,s}^{ES} - C_{e,t,s}^{ES}) - \gamma_{i,t,s}^P p_{i,t,s}^D = \sum_{j \in \mathcal{B}_i} p_{t,s}^{ij}, \quad \forall i \in \mathcal{B} \setminus \{1\}, \quad (3)$$

$$\sum_{u \in \mathcal{U}_i} q_{u,t,s}^{DG} + \sum_{e \in \mathcal{E}_i} q_{e,t,s}^{ES} - \gamma_{i,t,s}^P q_{i,t,s}^D = \sum_{j \in \mathcal{B}_i} q_{t,s}^{ij}, \quad \forall i \in \mathcal{B} \setminus \{1\}. \quad (4)$$

2) *Contingency Model*: The proposed model integrates conic relaxation of power flow equations in (6)–(14) to turn the original power flow problem into a convex second-order cone programming (SOCP) problem, which can be solved efficiently as compared to nonlinear and non-convex power flow equations that make the optimization problem NP-hard and does not guarantee optimality [52]. Specifically, power flow is obtained by applying conic relaxation on branch flow formulations, which is realized by introducing auxiliary variables $\nu_{i,t,s}, R_{t,s}^{ij}, T_{t,s}^{ij}$, expressed as:

$$\begin{aligned} \nu_{i,t,s} &= v_{i,t,s}^2 / \sqrt{2}, \\ R_{t,s}^{ij} &= v_{i,t,s} v_{j,t,s} \cos(\theta_{i,t,s} - \theta_{j,t,s}), \\ T_{t,s}^{ij} &= v_{i,t,s} v_{j,t,s} \sin(\theta_{i,t,s} - \theta_{j,t,s}), \end{aligned} \quad (5)$$

where $v_{i,t,s}$ and $\theta_{i,t,s}$ are the voltage magnitude and phase angle of bus i , respectively. Active and reactive power flow of lines are calculated in (6) and (7), where g^{ij} and b^{ij} are the conductance and the susceptance of line section ij , respectively. Auxiliary variables $u_{i,t,s}^{ij}, u_{j,t,s}^{ij}$ in constraints (8)–(11) define the behavior of voltages in adjacent buses connected through line $ij \in \mathcal{L}$, according to the line status at time t of scenario s , shown by $\alpha_{t,s}^{ij}$. When the line is connected, $\alpha_{t,s}^{ij} = 1$, and $u_{i,t,s}^{ij}, u_{j,t,s}^{ij}$ are equal to $\nu_{i,t,s}, \nu_{j,t,s}$. If the line is disconnected, then $\alpha_{t,s}^{ij} = 0$, setting $u_{i,t,s}^{ij}, u_{j,t,s}^{ij}$ to zero and forcing the power flow $p_{t,s}^{ij}, q_{t,s}^{ij}$ to zero in (12). Line power flow and voltage limit of lines are set in (13) and (14), respectively.

$$p_{t,s}^{ij} = \sqrt{2} g^{ij} u_{i,t,s}^{ij} - g^{ij} R_{t,s}^{ij} - b^{ij} T_{t,s}^{ij}, \quad \forall ij, \quad (6)$$

$$q_{t,s}^{ij} = -\sqrt{2} \left(b^{ij} + \frac{b_{sh}^{ij}}{2} \right) u_{i,t,s}^{ij} + b^{ij} R_{t,s}^{ij} - g^{ij} T_{t,s}^{ij}, \quad \forall ij, \quad (7)$$

$$0 \leq u_{i,t,s}^{ij} \leq \frac{\bar{v}_i^2}{\sqrt{2}} \alpha_{t,s}^{ij}, \quad \forall ij, \quad (8)$$

$$0 \leq u_{j,t,s}^{ij} \leq \frac{\bar{v}_j^2}{\sqrt{2}} \alpha_{t,s}^{ij}, \quad \forall ij, \quad (9)$$

$$0 \leq \nu_{i,t,s} - u_{i,t,s}^{ij} \leq \frac{\bar{v}_i^2}{\sqrt{2}} (1 - \alpha_{t,s}^{ij}), \quad \forall ij, \quad (10)$$

$$0 \leq \nu_{j,t,s} - u_{j,t,s}^{ij} \leq \frac{\bar{v}_j^2}{\sqrt{2}} (1 - \alpha_{t,s}^{ij}), \quad \forall ij, \quad (11)$$

$$2u_{i,t,s}^{ij} u_{j,t,s}^{ij} \geq \left(R_{t,s}^{ij} \right)^2 + \left(T_{t,s}^{ij} \right)^2, \quad R_{t,s}^{ij} \geq 0, \quad \forall ij, \quad (12)$$

$$\left(p_{t,s}^{ij} \right)^2 + \left(q_{t,s}^{ij} \right)^2 \leq \left(\bar{S}^{ij} \right)^2, \quad \forall ij, \quad (13)$$

$$\frac{v_i^2}{\sqrt{2}} \leq \nu_{i,t,s} \leq \frac{\bar{v}_i^2}{\sqrt{2}}, \quad \forall i. \quad (14)$$

3) *Distributed Generator Model*: The active and reactive power supplied by DGs are constrained in (15)–(16), where $\underline{p}_u^{DG}, \underline{q}_u^{DG}, \bar{q}_u^{DG}$ are the maximum active power, minimum reactive power, and the maximum reactive power limits of DG u , respectively. The total power generation of DGs during $\mathcal{T}'$ is limited in (17) to the available fuel, shown by $\bar{F}_u$.

$$0 \leq p_{u,t,s}^{DG} \leq \bar{p}_u^{DG}, \quad \forall u, \quad (15)$$

$$\underline{q}_u^{DG} \leq q_{u,t,s}^{DG} \leq \bar{q}_u^{DG}, \quad \forall u, \quad (16)$$

$$\sum_{t \in \mathcal{T}'} p_{u,t,s}^{DG} \Delta t \leq \bar{F}_u, \quad \forall u. \quad (17)$$

4) *Energy Storage System Model*: Constraints (18)–(21) model the operation of ES in power distribution system. The state equation of ES systems is formulated in (18), where charging and discharging dynamically changes the state-of-charge of ES systems, shown by $E_{e,t,s}$. The state-of-charge is limited in (19), to the minimum required storage $\underline{E}_e$ and the maximum storage capacity $\bar{E}_e$. The charging and discharging power of ES systems are bounded to their maximum limit $\bar{B}_e$ in (20). The reactive power limits of ES systems are defined in (21), where $\underline{q}_e^{ES}, \bar{q}_e^{ES}$ are the lower and upper bounds of reactive power, respectively.

$$E_{e,t+1,s} = E_{e,t,s} + (\eta_e^c C_{e,t,s}^{ES} - D_{e,t,s}^{ES} / \eta_e^d) \Delta t, \quad \forall e, \forall t \in \mathcal{T}' \setminus \{T\}, \quad (18)$$

$$\underline{E}_e \leq E_{e,t,s} \leq \bar{E}_e, \quad \forall e, \quad (19)$$

$$0 \leq C_{e,t,s}^{ES}, D_{e,t,s}^{ES} \leq \bar{B}_e, \quad \forall e, \quad (20)$$

$$\underline{q}_e^{ES} \leq q_{e,t,s}^{ES} \leq \bar{q}_e^{ES}, \quad \forall e. \quad (21)$$

C. Cascading Water Shortage Analysis Model

The proposed model for cascading water shortage analysis in WDS caused by the PDS power outage is presented here. This model takes the results of PDS outage analysis model as input to specify the status of water pumps in each scenario, i.e., impacted by power outage or not. The proposed model extends the formulation in [53] to consider the pumps' status after the power outage, represented by their capability to operate

with the available PDS power supply and the water demand shortage required at nodes with limited water supply. The model, applicable for both radial and meshed WDSs, adapts the WDS operation to minimize the impact of power outage in the water supply, by using the available pumps and stored water in tanks to maintain appropriate head levels and water flow along the pipelines and by controlling the amount of water supplied to certain loads to maintain the water balance. More specifically, objective function of the proposed model minimizes the cost of water not supplied during each sub-horizon $\mathcal{T}'$ in outage scenarios s as follows:

$$\min \sum_{t \in \mathcal{T}'} \sum_{n \in \mathcal{N}} WC_n \times (1 - \gamma_{n,t,s}^W) W_{n,t,s}^D \Delta t, \quad (22)$$

where $W_{n,t,s}^D$ is the water demand at node n at time t defined over the set of WDS nodes $\mathcal{N} = \{1, \dots, N\}$, $\gamma_{n,t,s}^W$ is the ratio of water served in node n at time t in scenario s , and WC_n is the cost of water not supplied per cubic meter in node n . The cost coefficients are assumed as fixed for each water demand node that are influenced by the geographical location of the nodes [54]. The proposed model determines the ratio of water supplied in each node—or conversely the water not supplied in each node as shown in (22) by $(1 - \gamma_{n,t,s}^W)$ —such that the cost of water not supplied is minimized. The objective function (22) is subject to constraints (23)–(39) defined for all $t \in \mathcal{T}'$, except where indicated, which include the hydraulic constraints of pipelines and tanks, as well as pumps' status after a power outage.

1) *Water Mass Balance*: The nodal water balance constraint is expressed in (23), where $\mathbf{W}_{t,s}^R$ is a $N \times 1$ vector of water injections from reservoirs to water nodes, $\mathbf{A}$ is a $P \times N$ pipeline-to-node incidence matrix mapping the pipeline connections to the water nodes, $\mathbf{Q}_{t,s}$ is a $P \times 1$ vector of pipeline water flow, and $\mathbf{C}_{t,s}, \mathbf{D}_{t,s}$ are $U \times 1$ vectors of water tank charging and discharging flows, respectively. The ratio of water supplied in nodes, $\gamma_{n,t,s}^W$, form variables vector $\mathbf{\Gamma}_{t,s}^W = [\gamma_{1,t,s}^W, \dots, \gamma_{N,t,s}^W]^T$ that is bounded in (24).

$$\mathbf{W}_{t,s}^R - \mathbf{\Gamma}_{t,s}^W \odot \mathbf{W}_{t,s}^D = \mathbf{A} \mathbf{Q}_{t,s} + \mathbf{A}^T [\mathbf{C}_{t,s} - \mathbf{D}_{t,s}], \quad (23)$$

$$\mathbf{0} \leq \mathbf{\Gamma}_{t,s}^W \leq \mathbf{1}_N. \quad (24)$$

2) *Tank Operation Model*: The state equation of tanks is formulated in (25), which tracks the available water in tanks, $\mathbf{V}_{t,s}$, given the water input and output of the tanks. The volume of water stored in a tank is also a function of the pressure head $\mathbf{H}_{t,s}$ at the connection node and size of the tank $\mathbf{\Delta}$ in (26). The limits of water stored in tanks and the initial volume in tanks are set in (27), where $\bar{\mathbf{V}}, \underline{\mathbf{V}}$ are the vectors of maximum and minimum tank capacity, and $\mathbf{V}^{init}$ is the vector of initial water volume. The input and output flow limits of tanks are set in (28)–(29), where $\bar{\mathbf{D}}, \bar{\mathbf{C}}$ are vectors of maximum input and output rates.

$$\mathbf{V}_{t+1,s} = \mathbf{V}_{t,s} + (\mathbf{C}_{t,s} - \mathbf{D}_{t,s}) \Delta t, \quad \forall t \in \mathcal{T}' \setminus \{T\}, \quad (25)$$

$$\mathbf{V}_{t,s} = \mathbf{\Delta} \mathbf{H}_{t,s}, \quad (26)$$

$$\underline{\mathbf{V}} \leq \mathbf{V}_{t,s} \leq \bar{\mathbf{V}}, \quad \mathbf{V}_{1,s} = \mathbf{V}^{init}, \quad (27)$$

$$\mathbf{0} \leq \mathbf{D}_{t,s} \leq \bar{\mathbf{D}}, \quad (28)$$

$$\mathbf{0} \leq \mathbf{C}_{t,s} \leq \bar{\mathbf{C}}. \quad (29)$$

3) *Water Pipeline Model*: The pipeline head losses are expressed using the Hazen-Williams equation in (30), where $\mathbf{G}$ is a $P \times P$ diagonal matrix of friction coefficients of pipes, and $\Pi(\cdot)$ is a sign function. The pipeline water flow is bounded by the maximum water flow vector $\bar{\mathbf{Q}}$ (31). The hydraulic head of nodes is constrained between the minimum and maximum limits $\underline{\mathbf{H}}$ and $\bar{\mathbf{H}}$ in (32). The head for water reservoirs is fixed to the elevation at the reservoir location in (33), where $\mathbf{\Theta}$ is a matrix mapping the pressure head at water nodes to the reservoirs' elevation $\mathbf{Z}$.

$$\mathbf{A}^T \mathbf{H}_{t,s} = \mathbf{G} \Pi(\mathbf{Q}_{t,s}) |\mathbf{Q}_{t,s}|^{1.852}, \quad (30)$$

$$\mathbf{0} \leq \mathbf{Q}_{t,s} \leq \bar{\mathbf{Q}}, \quad (31)$$

$$\underline{\mathbf{H}} \leq \mathbf{H}_{t,s} \leq \bar{\mathbf{H}}, \quad (32)$$

$$\mathbf{\Theta} \mathbf{H}_{t,s} = \mathbf{Z}. \quad (33)$$

4) *Variable-Speed Pump Model*: The head rise created by pumps, $\Delta \mathbf{H}_{t,s}^P$, is expressed as a function of the water flow and the pump speed $\mathbf{\Omega}_{t,s}$ in (34), where a, b, c are the characteristic parameters of M pumps. The pump head rise is related with the water nodes connected to the pumps in (35). The power consumption of pumps, which is supplied from the power distribution system $\mathbf{P}_{t,s}^W$ and the pump backup generators $\mathbf{P}_{t,s}^B$, is calculated in (36) as a function of the water flow in pumps $\mathbf{Q}_{t,s}^P$. The pump power consumption supplied by the grid is bounded by the maximum power $\bar{\mathbf{P}}^W$ in (37), where the binary parameter vector $\mathbf{I}_{t,s}$ expresses the power outage status of pumps, and is equal to zero when the pump has a power outage in scenario s , fixing the power withdrawn from the power distribution system to zero. The power output from the backup generators is limited to its maximum capacity $\bar{\mathbf{P}}^B$ when the power outage status is zero in (38) or forced to zero if the power supply for the pump is enabled. The energy consumed in the horizon $\mathcal{T}'$ must not exceed the maximum available energy of the backup generators $\bar{\mathbf{F}}^B$, as given by (39).

$$\Delta \mathbf{H}_{t,s}^P = \mathbf{\Omega}_{t,s}^2 \odot [a \mathbf{1}_M - b((\mathbf{1}_M \odot \mathbf{\Omega}_{t,s}) \odot \mathbf{B} \mathbf{Q}_{t,s})^c], \quad (34)$$

$$\Delta \mathbf{H}_{t,s}^P = \mathbf{B} \mathbf{A}^T \mathbf{H}_{t,s}, \quad (35)$$

$$\mathbf{P}_{t,s}^W + \mathbf{P}_{t,s}^B = \frac{\rho g}{\eta_{pump}} \Delta \mathbf{H}_{t,s}^P \odot \mathbf{Q}_{t,s}^P, \quad (36)$$

$$\mathbf{0} \leq \mathbf{P}_{t,s}^W \leq \bar{\mathbf{P}}^W \odot \mathbf{I}_{t,s}, \quad (37)$$

$$\mathbf{0} \leq \mathbf{P}_{t,s}^B \leq \bar{\mathbf{P}}^B \odot (\mathbf{1}_M - \mathbf{I}_{t,s}), \quad (38)$$

$$\sum_{t \in \mathcal{T}'} \mathbf{P}_{t,s}^B \Delta t \leq \bar{\mathbf{F}}^B. \quad (39)$$

IV. RESILIENCE METRICS FOR INTERDEPENDENT POWER AND WATER DISTRIBUTION SYSTEMS

This section presents the proposed metrics to quantify the interdependence between resilience of power and water distribution systems. The proposed metrics take as input the spatio-temporal power outage in PDS and the resulting cascading water

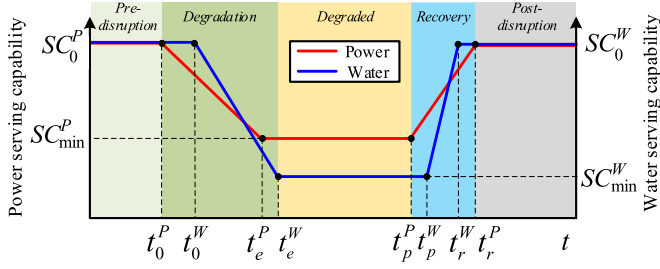


Fig. 3. Evolution of power and water serving capability in time.

shortage in WDS during outages, in terms of the ratio of power served, $\gamma_{i,t,s}^P$ and the ratio of water served, $\gamma_{n,t,s}^W$, respectively for PDS buses $i \in \mathcal{B}$ and WDS nodes $n \in \mathcal{N}$, for the set of outage scenarios $\mathcal{S}$ during $\mathcal{T}$, which are calculated using the PDS and WDS outage analysis models proposed in Section III. We first define two metrics, *Power Serving Capability*, shown by $SC^P(t)$, and *Water Serving Capability*, shown by $SC^W(t)$, as the average ratio of power and water served respectively by the PDS and WDS in $\mathcal{S}$ outage scenarios during $\mathcal{T}$, as follows:

$$SC^P(t) = \frac{1}{B \cdot S} \sum_{s \in \mathcal{S}} \sum_{i \in \mathcal{B}} \gamma_{i,t,s}^P, \quad (40)$$

$$SC^W(t) = \frac{1}{N \cdot S} \sum_{s \in \mathcal{S}} \sum_{n \in \mathcal{N}} \gamma_{n,t,s}^W. \quad (41)$$

The evolution of $SC^P(t)$ during outages and its cascading impact in the WDS operation, shown by $SC^W(t)$, is divided into multiple phases spanning the pre-disruption to post-disruption phases, as hypothetically illustrated in Fig. 3. In Fig. 3, the outage in the PDS occurs at t_0^P . The impact of the power outage on WDS, in terms of water shortage, starts later at t_0^W , depending on the outage location and available water stored in tanks. The PDS experiences a decay from its initial state SC_0^P to its degraded condition $SC_{\min}^P$ at time t_e^P , where the external disruption subsides and the repair crew actions to restore the PDS operation are scheduled. Restoration of PDS faulted lines starts at time t_p^P . The degraded state of the WDS operation, $SC_{\min}^W$, could be reached in a different time t_e^W , since the degradation from its initial state SC_0^W is dependent on the discharge flow rate of water tanks. A similar condition is experienced during the restoration, where WDS restoration may start after PDS restoration, at time t_p^W in Fig. 3. Power restoration is expected to be completed at time t_r^P , while water recovery may be achieved before or after at time t_r^W , depending on the tank water volume and restoration of line sections supplying power for pumping. Based on the evolving performance of PDS and WDS in supplying power and water to the customers, quantified using $SC^P(t)$ and $SC^W(t)$, the following metrics are introduced to quantify the interdependence between the resilience of power and water infrastructures.

1) *WDS Sensitivity to PDS Degradation (R_D)*: This metric establishes a relationship between WDS and PDS performances during the degradation stage by comparing the rates of decay from the pre-disruption stage to the degraded stage of each

system as follows:

$$R_D = \frac{SC_0^W - SC_{\min}^W}{SC_0^P - SC_{\min}^P} \times \frac{t_0^P - t_e^P}{t_0^W - t_e^W}. \quad (42)$$

The comparison of rates of decay establishes a measure of how sensitive the WDS response is to the disruptions in the PDS during the degradation stage. Hence, larger values of R_D show an increased dependence of WDS on the PDS operation.

2) *WDS Recovery Dependence to PDS Restoration (R_Q)*: This metric determines the recovery rapidity of WDS operation compared to the PDS restoration rate when transitioning from the degraded stage to the post-disruption stage and is calculated as follows:

$$R_Q = \frac{SC_0^W - SC_{\min}^W}{SC_0^P - SC_{\min}^P} \times \frac{t_r^P - t_p^P}{t_r^W - t_p^W}. \quad (43)$$

The value of R_Q measures the sensitivity of how fast WDS recovers once the PDS is restored, indicating the need for power restoration to ensure the WDS operation.

3) *Relative Impact of PDS Disruptions on WDS Operation (R_A)*: This metric measures the relative impact of outages in the PDS and the resulting cascading outages in the WDS operation, which is calculated as follows:

$$R_A = \frac{\sum_{t \in \mathcal{T}} SC^W(t)}{\sum_{t \in \mathcal{T}} SC^P(t)}. \quad (44)$$

The R_A values lower than one represent a higher degradation on the WDS compared to the PDS, and values larger than one represent a less dependent operation of the WDS to meet the water demand despite the PDS degradation.

4) *Dependence of WDS Water Shortage to PDS Power Outage (I)*: This metric is based on the mathematical definition of cross-correlation, which provides a measure of the association between the power and water serving capability metrics, $SC^P(t)$ and $SC^W(t)$:

$$I = \max_{\tau} \left\{ \sum_{t=-\infty}^{+\infty} (1 - SC^P(t))(1 - SC^W(t + \tau)) \right\}, \quad (45)$$

where larger values of I indicate increased dependency of the WDS operation to PDS disruptions. The formulation can be extended to define the dependence of each WDS node shortage to PDS power outages. Let $SC_n^W(t)$ be the water serving capability of demand n at time t :

$$SC_n^W(t) = \frac{1}{S} \sum_{s \in \mathcal{S}} \gamma_{n,t,s}^W. \quad (46)$$

The dependence of the water demand at node n to the power outages, I_n , is given by:

$$I_n = \max_{\tau} \left\{ \sum_{t=-\infty}^{+\infty} (1 - SC^P(t))(1 - SC_n^W(t + \tau)) \right\}. \quad (47)$$

5) *Expected Delay for WDS Degradation (Δt_D)*: The WDS operation could be sustained for a duration after the occurrence of the PDS disruption, either because the outage locations are not affecting the WDS operation, or the water demand is supplied from the water stored in tanks. This metric measures the period

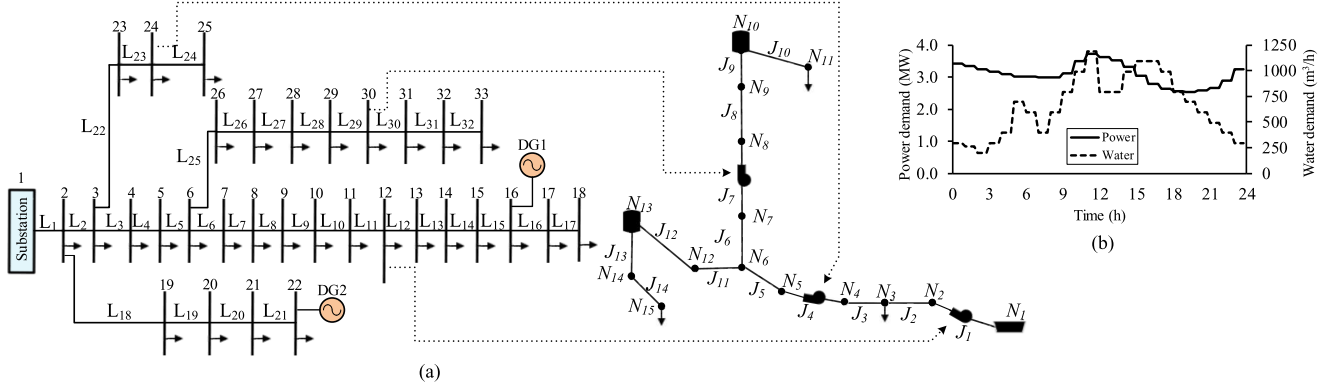


Fig. 4. (a) interdependent test 33-bus power and 15-node water distribution systems, (b) total power and water demand profiles for the systems.

that a WDS sustains the operation during the power outage, which is defined as follows:

$$\Delta t_D = t_0^W - t_0^P. \quad (48)$$

6) *Expected Delay of WDS Recovery (Δt_Q)*: This metric determines the difference between the recovery time of WDS and PDS, which may be different given the outage locations, the WDS connections to the PDS, and the WDS ability to supply the water demand as the pump operation is restored. This metric is defined as follows:

$$\Delta t_Q = t_r^P - t_r^W. \quad (49)$$

Negative values of Δt_Q indicate a restoration delay in the WDS. Positive values show that the dependence between the two systems is low, indicating that the total restoration of the PDS is not necessary to fully restore the WDS operation.

V. NUMERICAL STUDY

The proposed resilience analytics model for interdependent power and water distribution systems is utilized in this section for resilience assessment of a 15-node WDS interconnected with the IEEE 33-bus test power distribution system [2], which is subject to high speed wind that causes power outage in the test PDS, and results in cascading outages in the test WDS. The diagram of interdependent test PDS and WDS is shown in Fig. 4-(a), where the pumps J1, J4, and J7 are supplied by buses 12, 24, and 30 of the test PDS, respectively. Two water tanks with maximum capacity $1,200 \text{ m}^3$ are operated at nodes 10 and 13 of the test WDS, with initial stored water of respectively $1,000 \text{ m}^3$ and 500 m^3 . Two controllable distributed generators are also available in buses 16 and 22 of the test PDS. The power and water demand profiles are shown in Fig. 4-(b), with peak values of 3.72 MW and $1,188 \text{ m}^3/\text{h}$ for the PDS and WDS, respectively. The power profiles are obtained from CAISO for March 29–30, 2017, and the rest of the data for the 15-node WDS is available at [55].

For the spatio-temporal outage scenario generation, the set of line outages at each time step $t \in \mathcal{T}$ is determined by initially calculating the failure probability $\pi_{l,t}^v$ of each line $l \in \mathcal{L}$ for a wind speed v . The line failure probability is modeled as a function of the wind speed at the line location using piecewise

fragility curves [18], [20]. The wind speed is represented by a Gaussian function scaled to a maximum wind speed of 78 mph , with mean value equal to the expected disruption time, set for hour 12, and a standard deviation of 2 hours. Once the failure probability $\pi_{l,t}^v$ is determined, a uniformly distributed random number $r_{l,t}$ is generated for each line l at time t and is compared to $\pi_{l,t}^v$ to determine the line status. If $r_{l,t} \leq \pi_{l,t}^v$, the line is tripped starting from t until the specified restoration time; otherwise, the line operation remains in its normal condition. The restoration time is sampled from a normal distribution with a mean value equal to the mean-time-to-repair, which is assumed as 6 hours with a standard deviation of 1 h for all lines.

Seven cases are studied for outage assessment and resilience metrics evaluation of the test interdependent PDS and WDS in Fig. 4. These cases consider different combinations of realistic upgrades in the power and water distribution systems, represented by the inclusion of distributed power sources at different locations and the increase of water storage capacity in tanks, respectively. In Case 0, the operation of test PDS and WDS is analyzed under 20 spatio-temporal single-equipment outage scenarios, without DGs in PDS and restricting the water storage in WDS tanks. Case 1 studies the operation of the systems under the same scenarios in Case 0, with increased tank capacity of $1,200 \text{ m}^3$ and initial volume in tanks 10 and 13 of $1,000 \text{ m}^3$ and 500 m^3 , respectively. Case 2 extends the analysis of the effect of the water storage to enhance the WDS resilience considering a tank capacity of $2,500 \text{ m}^3$ and initial volume $2,200 \text{ m}^3$ and $1,900 \text{ m}^3$, respectively. In Case 3, the response of the systems to more severe disturbances is analyzed, where scenarios may contain multiple outages to occur during the study window. Case 4 studies the effect of DGs on the resilience of the interdependent PDS and WDS, while other conditions are the same as Case 3. Cases 5 and 6 evaluate the impact of DG location on the interdependence between the resilience of power and water distribution systems. Case 5 considers the severe scenarios evaluated for Cases 3 and 4, when DGs are located at power buses 22 and 33, far from the pumps' supply points. Case 6 considers DGs connected at buses 10 and 25, which are closer to the supply points of pumps J1 and J4 from the power distribution grid. In Cases 3–6, a set of 20 spatio-temporal single and multiple-equipment outage scenarios are simulated. A summary of the cases, including the type of

TABLE I
SUMMARY OF CASE STUDIES

Case	Type of outage scenario	DG available in PDS?	DG location in PDS	Tank volume
0	Single	No	—	None
1	Single	No	—	Low
2	Single	No	—	High
3	Multiple	No	—	High
4	Multiple	Yes	base location	High
5	Multiple	Yes	far from pumps	High
6	Multiple	Yes	close to pumps	High

TABLE II
RESILIENCE METRICS COMPARISON FOR THE ANALYZED CASES

Case	R_D	R_Q	R_A	I	$\Delta t_D(h)$	$\Delta t_Q(h)$
0	1.753	3.964	0.933	2.899	0	1
1	1.01	3.812	0.957	2.455	0	1
2	0.68	2.601	0.987	1.876	1	1
3	1.614	1.633	0.959	4.771	2	0
4	0.617	1.072	1.034	1.59	2	0
5	1.43	2.221	0.985	2.835	2	1
6	0.352	0.337	1.087	0.534	2	0

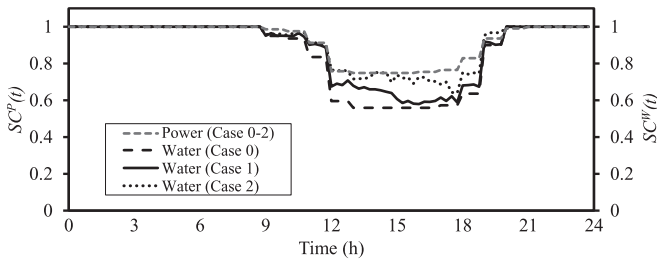


Fig. 5. Comparison of power and water serving capability in Cases 0, 1, 2.

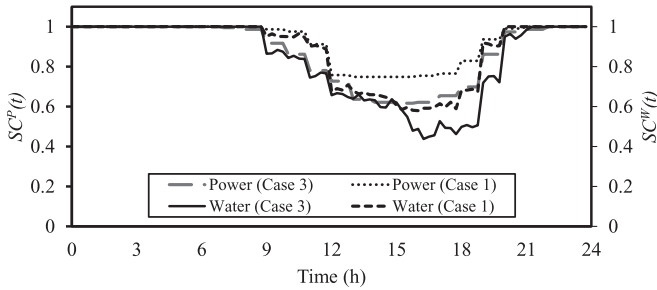


Fig. 6. Comparison of power and water serving capability in Cases 1 and 3.

outages considered, along with the power and water distribution system enhancements is presented in Table I. Note that Cases 0–6 do not consider the availability of backup generation in pumping stations, which is studied later in Section V-C.

The resilience metrics for all cases are summarized in Table II. Further, the temporal evolution of power and water load serving capability metrics, $SC^P(t)$ and $SC^W(t)$, of the test PDS and WDS in Cases 0–2 are shown in Fig. 5. In addition, the serving capability comparison between Cases 3 and 1 is presented in Fig. 6 to evaluate the impacts of the outage severity. In Cases 0 and 1, the WDS performance decays immediately when the PDS

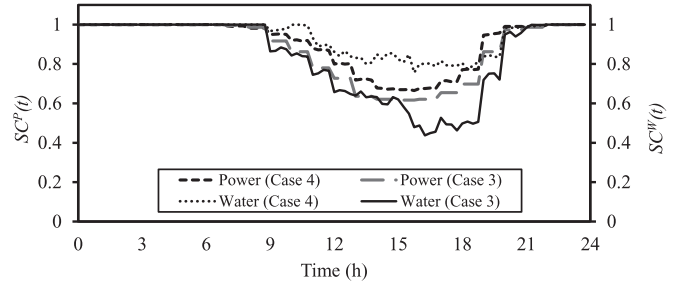


Fig. 7. Comparison of power and water serving capability in Cases 3 and 4.

outage occurs at hour 9. In Case 2, where WDS has more than twice the water tank capacity compared to Case 1, the WDS has maintained water supply for a longer duration after the power outage in pumps. The increased water tank capacity in Case 2 also leads to a slower rate of decay compared to Cases 0 and 1, as evidenced from the respective 61.2% and 32.6% reduction in R_D metric in Table II, since the water demand can be supplied for a longer period compared to Cases 0 and 1, reducing the rate of decay after the PDS disruption. When the power serving capability of PDS reaches its minimum value at hour 13 in Fig. 5 along with the water serving capability of WDS in Case 0, WDS degradation continues until time 15:45 for Case 1 and 17:45 for Case 2, when initial restoration actions are applied. During the recovery stage, the WDS with larger water tank capacity in Case 2 has a reduction of 31.7% and 35.7% of R_Q metric in Table II compared to Cases 1 and 0, respectively, which indicates lower dependence on power restoration for WDS in Case 2. In both Cases 1 and 2, the restoration of WDS operation is achieved 1 h before the PDS restores according to Δt_Q in Table II, indicating that the restoration of some lines does not have a direct impact in the WDS operation. The relative impact given by metric R_A indicates that the PDS disruption was less critical for Case 2, as expected, showing a more resilient operation with enhanced robustness due to the larger water tank capacity. In general, the ability of the WDS to store water in tanks directly influences the WDS resilience, by sustaining the water supply when power outages cause interruptions in the flow from water sources.

In Case 3, where the interdependent PDS and WDS operation is analyzed under multiple-outage scenarios, the power serving capability of PDS decays to a minimum of 0.617 at hour 13 in Fig. 6, which is lower than the minimum value of Case 1, due to the increased disturbance caused by multiple-outage scenarios. In Case 3, water demand is supplied for 2 hours after the power outages occur according to Δt_D in Table II, followed by 5:45 decay from the normal operating state before reaching the minimum water serving capability of 0.437. Comparing R_D metric for Case 1 and 3 indicates an increased 59.8% WDS sensitivity to PDS disruptions, caused by the increased number of lines outages, which is also indicated by the relative impact metric R_A in Table II.

The power and water serving capability of the systems in Case 4, where DG is added to the test PDS, is shown and compared with Case 3 in Fig. 7. In Fig. 7, the power curtailment in Case 4 is reduced as compared to Case 3, since DGs supply power

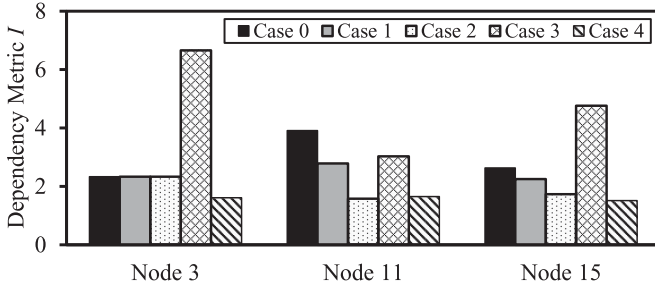


Fig. 8. Dependency of water shortage in WDS nodes to power outages.

to the line sections isolated from the main substation during the outages. The WDS operation also benefits from the inclusion of the DGs. In Case 4, the availability of power from DGs enhances the WDS operation by reducing the pump power shortages and the resulting water curtailment, with a reduced relative impact compared to the PDS as given by R_A in Table II. In Table II, the WDS sensitivity to PDS disruptions, R_D , is reduced by 61.7% compared to Case 3, as additional power resources reduce the degradation rate of water supply. In addition, the extra resources reduce the dependence of the WDS to the PDS restoration as measured by R_Q in Table II.

The impact of the DG location is discussed based on the results obtained from Cases 5 and 6, presented in Table II. In Case 5, the location of DGs is far from the WDS supply points, which represents a 131.7% increase in the WDS sensitivity to PDS degradation, compared to the DG configuration proposed in Case 4. Although it represents an improvement compared to the case with no DG (Case 3), the WDS operation is more affected by the limitation to access backup power from the distributed resources, resulting in an increased dependency with the PDS restoration represented by a 78.3% increase in the metric I compared to Case 4. In Case 6, the dependence of water shortage to power outages, I , where DGs are conveniently located close to the WDS supply points of pumps J1 and J4, is reduced by 66.4% compared to Case 4. In this case, the sensitivity to the PDS degradation and dependence to the PDS restoration are reduced compared to Case 4, showing that the WDS is capable of operating with the available power sources, ensuring the water supply despite the outage occurrence. This implies that the available DGs and their proximity to the supply points between power and water distribution systems have impact on the WDS resilience, as DG units close to the WDS pumps provide backup generation, which ensures the pump operation and the water supply continuity.

A spatial analysis of the dependency metric I_n can help measure the dependence of water shortage in specific WDS nodes to power outage. In this regard, the dependency metric I_n is evaluated for WDS nodes 3, 11, and 15 and is depicted in Fig. 8. In Fig. 8, the dependence of water shortage in the WDS nodes 11 and 15 to power outage is reduced as more water tank capacity is added to WDS, as seen in Cases 0 to 2. In addition, the inclusion of DGs in Case 4 enhances the PDS resilience, which leads to lower power outages and consequently reduces water shortage in WDS that resulted from power outages. Further, the dependency metric increases as more severe power outage

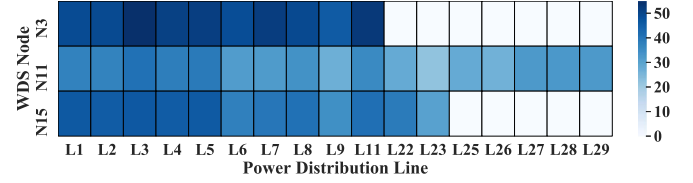


Fig. 9. Dependency of water shortage in WDS nodes to outage of individual PDS lines.

TABLE III
METRICS COMPARISON BASED ON THE NUMBER OF SCENARIOS

Number of scenarios		R_D	R_Q	R_A	I
10	Mean value	0.190	0.395	1.059	0.829
	Standard deviation	0.110	0.227	0.023	0.555
20	Mean value	0.269	0.528	1.045	0.747
	Standard deviation	0.104	0.190	0.018	0.371
50	Mean value	0.245	0.474	1.051	0.684
	Standard deviation	0.033	0.080	0.008	0.155
100	Mean value	0.245	0.461	1.055	0.708
	Standard deviation	0.029	0.054	0.007	0.112

scenarios happen in Case 3. More specifically, in node 11, since all the three pumps are required to supply the water demand, the node is very sensitive to power outages as evidenced by high value of I_n in Case 1, but the increased water tank capacity in Case 2 reduces its dependency to a power outage.

The dependency of water shortage in WDS nodes 3, 11, and 15 to the outage of individual lines in PDS is depicted in Fig. 9. The faults at PDS lines 1–9 and 11 have the most impact on water shortage in the three WDS nodes since they all cause the power outage in the reservoir pump J1. These contingencies expose a WDS vulnerability to the power outage in multiple power lines, which may need to be hardened to reduce the vulnerability. The faults at PDS line 25–29 do not cause power outage in pump J1 and therefore do not result in any water shortage in nodes 3 and 15, while causes a water shortage in node 11. Node 11 is the most vulnerable to power outage among the three WDS nodes, as the outage of many power distribution lines would result in water shortage in the node.

A. Impact of Number of Outage Scenarios

To assess the impact of the number of outage scenarios in the metrics' evaluation, the metrics were calculated considering 10, 20, 50, and 100 randomly drawn scenarios, based on the configuration of Case 3. This procedure is repeated 10 times for each number of scenarios, and the mean value and standard deviation obtained for each metric is shown in Table III. Based on the obtained results, the value of metrics R_D and R_Q varies 41.6% and 36.7% when scenarios are increased from 10 to 20, respectively, compared to increasing from 20 to 50 scenarios, where the variation reduces to 8.9% and 10.2%. Metric R_A shows a lower influence with respect to the number of scenarios, with a maximum deviation of 1.3% from 10 to 20 scenarios. On the other hand, metric I tends to reduce when increasing the number of scenarios. Metric I is based on the similarities between the power and water serving capabilities to recognize

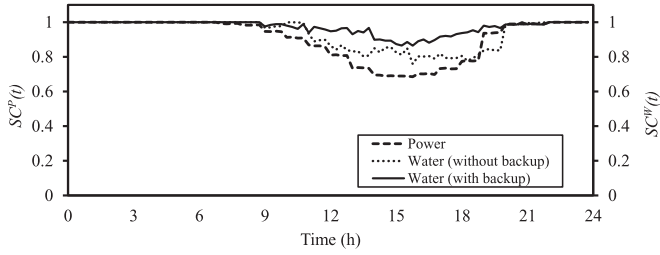


Fig. 10. Comparison of water serving capabilities with and without backup generation at pumping station J1.

TABLE IV
EFFECT OF PUMP BACKUP GENERATION ON THE RESILIENCE METRICS

Metric	R_D	R_Q	R_A	I	Δt_D	Δt_Q
No backup	0.617	1.072	1.034	1.590	2	0
With backup	0.467	0.404	1.071	0.679	2	0

patterns from the power serving capability that are reflected into the water serving capability. The effect of severe events, which are of interest for resilience assessment, might be attenuated while averaging a considerably high number of scenarios, which is reflected in the metric as a reduction in their interdependence. Note that these observations are made for the specific test systems in this session, and the results and variations may depend on the system under study.

B. Effect of Backup Generators Connected to Pumping Stations

As mentioned in Section II, urban WDSs are usually equipped with backup generators installed at pumping stations to support the pump operation during a power outage. To illustrate the benefits of the backup generation in the interdependence between the resilience of power and water distribution systems, outage scenarios of Case 4 are re-evaluated considering the presence of backup generation at pumping station J1, whose interruption compromises the water flow from the reservoir to all water demands. The power and water serving capabilities with and without backup generation are depicted in Fig. 10, along with the metrics in Table IV. The backup generator does not provide electricity to the power grid; hence, the power serving capability is the same for both cases. The results show that the water serving capability is increased when backup generation is available at pump J1, due to the ability to supply water to the demand connected at water node 3. The enhanced resilience is also captured by the proposed metrics. Reductions in metrics R_D and R_Q show that the WDS is less affected by the occurrence of power outages, and its recovery is less dependent on the PDS restoration when backup generators support the pump operation. The value of metric R_A is greater when backup generation is enabled, representing a lower degradation of the WDS operation from the power outages. The value of metric I is significantly lower compared to the case with no backup generation, showing that the backup generators ensure that the WDS water serving capability is less affected during power outages, as shown in

Fig. 10, representing a decreased dependency between the two infrastructures.

VI. CONCLUSION

This paper proposed an analytics model to evaluate the cascading effects of power outages on the operation of WDSs and to quantify the interdependence between the resilience of power and water distribution systems. The proposed analytics model includes an outage analysis model to determine the power serving capability of the PDS, and the resulting water serving capability of the WDS, in response to external events disrupting the grid operation. A set of six resilience metrics were defined to identify dependencies during the degradation and restoration stages, which are calculated using the power and water serving capabilities obtained from the outage analysis model. Numerical studies were carried out on the IEEE 33-bus test power distribution system, interconnected with a 15-node test water distribution system. The results indicate that the increased water tank capacity reduces the rate of decay of the water supply by compensating the amount of water shortage during the PDS outage. In addition, the allocation of DGs in PDS supplies power during outages, which reduces the power interruptions of pumps and the resulting water shortage in WDS. Although the availability of additional water tank capacity in WDS reduces the water shortage during power outages, the results indicate that the impact of options to enhance the resilience of PDS, such as utilizing DGs, is larger on enhancing the resilience of WDS to power outages. In addition, the use of backup generation weakens the coupling between the two infrastructures, mitigating the negative consequences of the power outages in the WDS operation. Based on these results, the proposed metrics provide a mechanism to evaluate potential upgrades in PDS and WDS infrastructures and quantify their benefits for the systems.

Future works include considering the impacts of other remedial actions by PDS, including network reconfiguration, in the proposed resilience analytics framework. Also, modeling the impacts of emerging technologies that intensify the interdependence of PDS on WDS, such as small pumped-storage hydropower, are of interest. Although it is expected that these technologies enhance the mutual resilience of both power and water infrastructures, as WDS can support the PDS and reduce power curtailments, the performance of the proposed metrics to capture this effect in a bidirectional interdependent operation needs to be assessed. Another potential research area is to develop the customer interruption cost function for water distribution facilities, which would provide the outage cost for a different water facility (such as pumping, desalination, and treatment) as a function of outage duration.

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